NOT MEASUREMENT SENSITIVE

MIL-PRF-46795H
29 October 2009
SUPERSEDING
MIL-PRF-46795G
29 August 1997

PERFORMANCE SPECIFICATION

GENERATING SYSTEM: ALTERNATOR, RECTIFIER, 60 AMPERES, 28 VOLTS

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers 28 volts (V), direct current (dc) generating systems, engine driven through belt or direct connection, to provide 60 amperes (A) for charging and operating loads. The system consists of a rotating field alternator with integral diode rectifier and adjustable voltage regulator in a single housing. This specification also covers repair parts for this system.

2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3 and 4 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirement documents cited in sections 3 and 4 of this specification, whether or not they are listed.

2.2 Government documents.

2.2.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract (see 6.2).

Comments, suggestions, or questions on this document should be addressed to U.S. Army RDECOM, Tank Automotive Research, Development and Engineering Center, ATTN: RDTA-EN/STND/TRANS MS #268, 6501 E. 11 Mile Road, Warren, MI 48397-5000 or emailed to DAMI_STANDARDIZATION@conus.army.mil. Since contact information can change, you may want to verify the currency of this address information using the ASSIST Online database at http://assist.daps.dla.mil.

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STANDARDS

DEPARTMENT OF DEFENSE STANDARDS

MIL-STD-461	-	Requirements for the Control of Electromagnetic Interference Characteristics of Subsystems and Equipment.
MIL-STD-810	-	Environmental Test Methods and Engineering Guidelines.
MIL-STD-889	-	Dissimilar Metals
MIL-STD-1275	-	Characteristics of 28 Volt DC Electrical Systems in Military Vehicles.

(Unless otherwise indicated, copies of the above specifications, standards, and handbooks are available from the Document Automation and Production Service, Building 4/D, 700 Robbins Avenue, Philadelphia, PA 19111-5094 or website: http://assist.daps.dla.mil/quicksearch/.)

2.2.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation or contract (see 6.2).

DRAWINGS

DEPARTMENT OF DEFENSE

10929868	-	Rectified AC Generating System (Interface).

(Copies of these drawings are available from the U.S. Army Tank-automotive and Armaments Command, ATTN: RDTA-EN, 6501 E. 11 Mile Rd, Warren, MI 48397-5000.)

NUCLEAR REGULATORY COMMISSION (NRC)

Code of Federal Regulations (CFR) - Title 10, Parts 30 and 40.

(Copies of the Code of Federal Regulations (CFR) are available from the Superintendent of Documents, U.S. Government Printing Office, 732 North Capitol St. NW, Washington, DC 20402 or website: http://www.gpoaccess.gov/cfr/.)

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2.3 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract (see 6.2).

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

ASTM B117 - Standard Practice for Operating Salt Spray (Fog) Apparatus (DoD Adopted).

(Application for copies should be addressed to the American Society for Testing and Materials (ASTM), 100 Barr Harbor Drive, West Conshohocken, PA 19428-2959 or website: http://www.astm.org.)

2.4 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First article. When specified (see 6.2), samples shall be subjected to first article inspection in accordance with 4.3.

3.2 Materials. Materials used shall be in accordance with the manufacturer's materials specifications for this type of item. The material shall be free of defects which adversely affect appearance, performance, or serviceability of the finished product (see 4.6.1).

3.2.1 Recycled, recovered, or environmentally preferable materials. Recycled, recovered, or environmentally preferable materials should be used to the maximum extent possible provided that the material meets or exceeds the operational and maintenance requirements, and promotes economically advantageous life cycle costs (see 4.6.1).

3.2.2 Prohibited material. Asbestos, cadmium, and radioactive materials shall not be used in this item. Radioactive material is defined by Title 10, CFR, Parts 30 and 40, and other radioactive material in which the specific activity is greater than 0.002 microcuries per gram or the activity per item equals or exceeds 0.01 microcuries (see 4.6.1).

3.2.3 Pure tin. The use of pure tin is prohibited. Tin content used in connector materials shall not exceed 97 percent, and an alloy material shall be chosen to inhibit the growth of tin whiskers (see 6.4.4).

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3.2.4 Dissimilar metals. When dissimilar metals are used in intimate contact with each other, protection against electrolysis and corrosion shall be provided. Dissimilar metals such as brass, copper or steel (except corrosion-resisting steel) passivated in accordance with SAE-AMS-2700 shall not be used in intimate contact with aluminum or aluminum alloy. Protective measures for dissimilar metals shall be in accordance with MIL-STD-889.

3.3 Design and construction. Generating system shall conform to the envelope dimensions, the mechanical mounting interfaces, and the electrical mounting interfaces as shown in drawing 10929868, and applicable specifications and standards (see 4.6.1).

3.3.1 Interchangeability. Generating system and all component parts shall be physically and functionally interchangeable with all other such parts of the same part number previously furnished under this specification (see 4.6.2).

3.3.2 Polarity. Generating systems shall be designed to operate in negatively grounded electrical circuit (see 4.6.2).

3.3.3 Standard parts. Military standard parts shall be incorporated whenever possible. Commercial standard parts may be used, provided they are interchangeable with military standard parts without modification and meet the requirements and verifications within this document (see 4.6.1).

3.4 Performance.

3.4.1 Electromagnetic interference characteristics. Conducted and radiated emissions from the generating system shall conform to requirements specified in figures 1 and 2 (see 4.7).

3.4.2 Voltage. Generating system shall provide steady state voltage at any setting between 26 volts direct current (Vdc) and 30 Vdc when mechanically and electrically installed and operated as intended (see 4.8).

3.4.2.1 Voltage adjustment. Generating system shall provide means to adjust voltage output between 26 Vdc and 30 Vdc (see 4.8.1).

3.4.2.2 Voltage regulation. Generating system, when output voltage is adjusted as in intended use, shall maintain set voltage (see 6.4.1) within $\pm$ 0.5 Vdc. Generating system shall return to set voltage within 0.6 second (s) of any load change (see 4.8.2).

3.4.2.3 Transient voltage characteristics. Generating system shall meet the requirements of MIL-STD-1275 regarding ripple, surges, spikes, and starting disturbances in fault-free, single fault, and multi-fault conditions (see 4.8.3).

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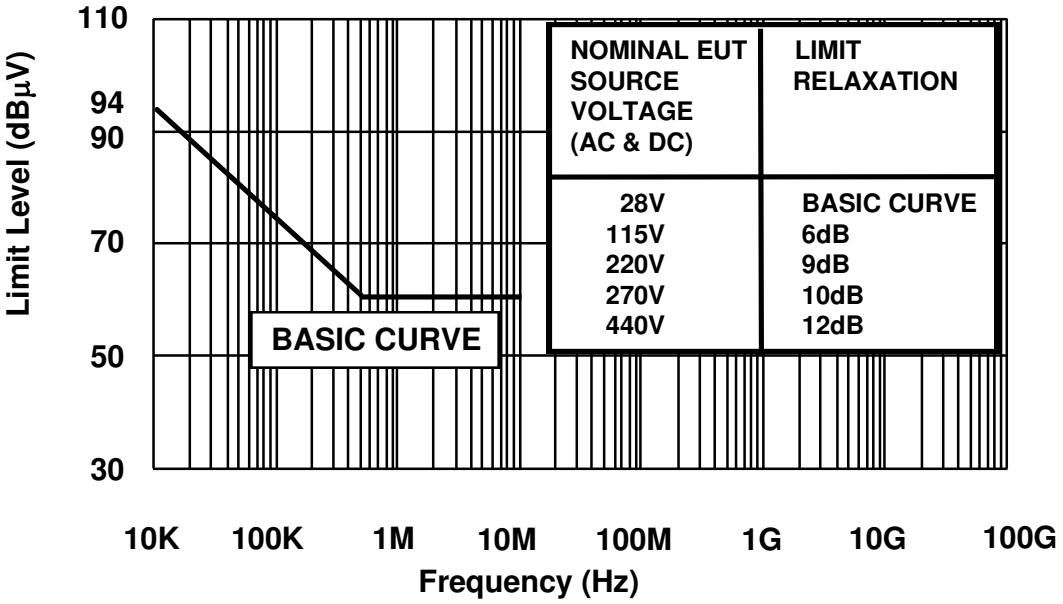


FIGURE 1. Conducted emissions levels.

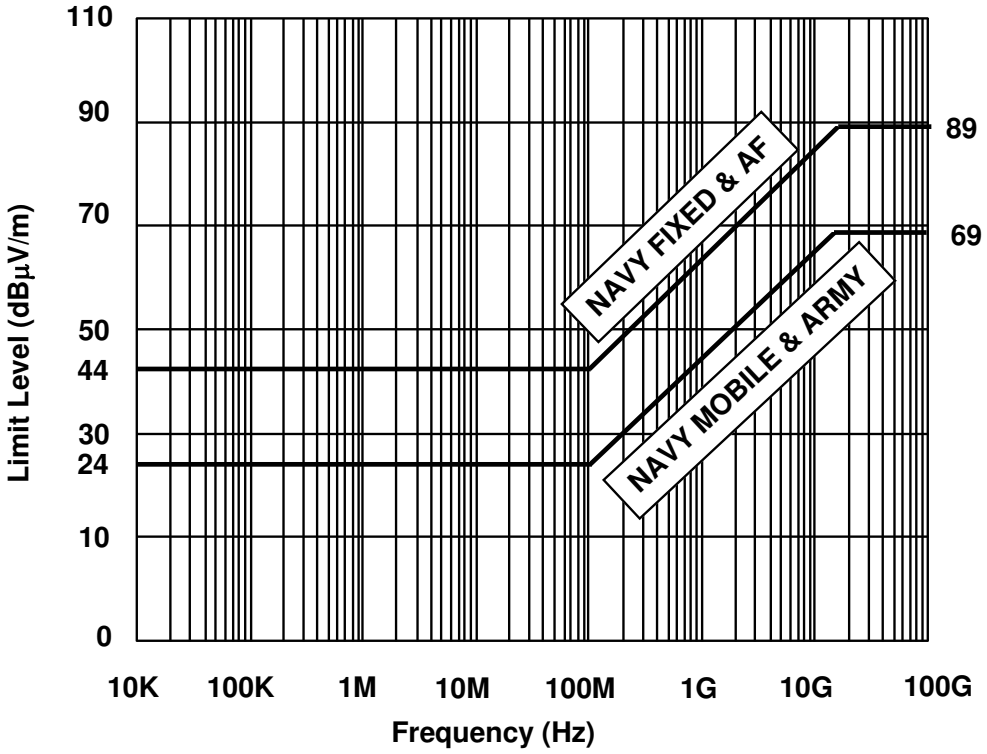


FIGURE 2. Radiated emissions levels.

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3.4.3 Maximum current (rated output). Generating system shall supply 60 A current under maximum load at 10 000 revolutions per minute (rpm) alternator rotor shaft speed (see 4.9).

3.4.3.1 Non-operating current. Generating system shall exhibit non-operating current through ignition switch circuit of not more than 300 milliamperes (mA). Further, with ignition switch open, non-operating current from battery shall be not more than 25 mA (see 4.9.2).

3.4.4 Endurance. Generating system shall maintain set voltage (see 3.4.2.2) during operation under various conditions of rotor shaft position, speed, and current load. There shall be no servicing, adjustment, nor replacement of parts (see 4.10).

3.4.5 Overspeed. Generating system shall operate at 12 000 rpm under not more than 5 A load without impairing electrical or mechanical performance (see 4.11).

3.4.6 High temperature operation. Generating system shall maintain set voltage output $\pm$ 0.2 Vdc during operation in ambient air temperature of 250 degrees Fahrenheit (°F) (121 degrees Celsius (°C)). Subsequently, system shall conform to 3.4.3 (see 4.12).

3.4.7 Low temperature operation. Generating system shall maintain set voltage output $\pm$ 0.5 Vdc during operation in ambient air temperature of minus (-) 65°F (-54°C). Subsequently, system shall conform to 3.4.3 (see 4.13).

3.4.8 Shock. After testing in accordance with 4.14, generating system shall meet the requirements of 3.4.2 and 3.4.3.

3.4.9 Vibration. During testing in accordance with 4.15, generating system shall maintain set voltage output $\pm$ 0.5 Vdc. After testing in accordance with 4.15, the generating system shall meet the requirements of 3.4.4.

3.4.10 Submersion. Generating system shall operate during submersion in a salt-water solution. Current from battery shall not exceed 10 A. Subsequently, voltage regulator compartment shall contain not more than 10 cubic centimeters (cm^3) of salt-water solution, and the system shall meet the requirements of 3.4.2 and 3.4.3 (see 4.16).

3.4.11 Fungus resistance. Generating system shall meet the requirements of 3.4.2 and 3.4.3 after testing in accordance with 4.17.

3.4.12 Corrosion resistance. Generating system shall meet the requirements of 3.4.2 and 3.4.3 after testing in accordance with see 4.18.

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3.4.13 Sand and dust. Generating system shall meet the requirements of 3.4.2 and 3.4.3 after testing in accordance with 4.19.

3.5 Finish. Unless otherwise specified (see 6.2), generating system shall be painted in accordance with the manufacturer's standard practice, with outer coat of black or olive drab enamel (see 6.3) on all exterior surfaces, except electrical connections and exposed shaft surfaces (see 4.6.2).

3.6 Identification of product.

3.6.1 Marking. Unless otherwise specified (see 6.2), generating system shall be marked in accordance with standard commercial practice (see 4.6.2).

3.6.2 Nameplate. Generating system shall have a nameplate attached to alternator frame in conspicuous location so that plate may be easily read when generating system is installed in vehicle. Nameplate shall contain the following minimum information (see 4.6.2):

GENERATING SYSTEM DC
60 A 28 VOLTS
MAXIMUM RATED SPEED - 10 000 RPM
SPECIFICATION MIL-PRF-46795
SERIAL NUMBER
NATIONAL STOCK NUMBER (NSN)
MANUFACTURER'S NAME OR IDENTIFICATION CODE
U.S.

3.7 Safety. The generating system shall be free from defects involving rotating parts, burrs or sharp edges, and electrical shock which may result in injury of personnel (see 4.6.2).

4. VERIFICATION

4.1 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.3).
b. Conformance inspection (see 4.4).
c. Control tests (see 4.5).

4.2 Inspection conditions. Unless otherwise specified (see 6.2), all inspections shall be conducted under the following standard (room) ambient conditions:

a. Temperature: 77 ± 15°F (25 ± 8.3°C).

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b. Relative humidity: Uncontrolled room ambient.
c. Atmospheric pressure: Site pressure.

4.2.1 Batteries. 24 Vdc power source shall be two 12 Vdc military type batteries connected in series. A battery which accepts a charging current of not more than 4.5 A at set voltage after storage at laboratory ambient air temperature shall be considered fully charged. In case of a high voltage reading, battery fluid specific gravity shall be inspected and, if over 1.30, battery shall be discharged until specific gravity is not more than 1.30, and charging current is not more than 4.5 A.

4.2.2 Stabilized frame temperature. Unless otherwise specified herein, generating system frame temperature shall be stabilized prior to undergoing test procedure (see 6.4.2).

4.2.2.1 Frame temperature. Frame temperature shall be measured at center of stator lamination stack on outside diameter.

4.2.3 Test voltage. Unless otherwise specified, voltage regulator shall be adjusted to 28 Vdc ± 0.2 Vdc with alternator running at 2000 rpm and producing not more than 10 A. This adjustment shall be made after frame temperature has been stabilized.

4.3 First article inspection. Samples of products to be furnished under this specification shall undergo the examinations in 4.4.2 and tests in order listed in table I (see 3.1).

TABLE I. Order of first article testing.

Sample number	Requirement	Test	Requirement
1	3.4.3	4.9.1	Rated output
	3.4.2.2	4.8.2	Voltage regulation
	3.4.3.1	4.9.2	Non-operating current
	3.4.4	4.10	Endurance
	3.4.3	4.9.1	Rated output
	3.4.2.2	4.8.2	Voltage regulation
2	3.4.3	4.9.1	Rated output
	3.4.2.2	4.8.2	Voltage regulation
1/	3.4.6	4.12	High temperature
1/	3.4.2.2	4.8.2	Voltage regulation
	3.4.3	4.9.1	Rated output
2/	3.4.7	4.13	Low temperature
2/	3.4.2.2	4.8.2	Voltage regulation
	3.4.3	4.9.1	Rated output
	3.4.8	4.14	Shock
3/	3.4.3	4.9.1	Rated output
3/	3.4.2.2	4.8.2	Voltage regulation

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TABLE I. Order of first article testing - Continued.

Sample number	Requirement	Test	Requirement
3	3.4.3	4.9.1	Rated output
	3.4.2.2	4.8.2	Voltage regulation
	3.4.5	4.11	Overspeed
	3.4.3	4.9.1	Rated output
4/	3.4.9	4.15	Vibration
4/	3.4.2.2	4.8.2	Voltage regulation
	3.4.3	4.9.1	Rated output
	3.4.2.2	4.8.2	Voltage regulation
	3.4.13	4.19	Sand and dust
	3.4.3	4.9.1	Rated output
	3.4.2.2	4.8.2	Voltage regulation
4	3.4.3	4.9.1	Rated output
	3.4.2.2	4.8.2	Voltage regulation
	3.4.10	4.16	Submersion
5/	3.4.3	4.9.1	Rated output
5/	3.4.2.2	4.8.2	Voltage regulation
5	3.4.3	4.9.1	Rated output
	3.4.2.2	4.8.2	Voltage regulation
	3.4.2.3	4.8.3.2	Load switching
	3.4.3	4.9.1	Rated output
	3.4.2.3	4.8.3.1	High voltage transients
	3.4.3	4.9.1	Rated output
6	3.4.12	4.18	Corrosion resistance (200 hours)
6/	3.4.3	4.9.1	Rated output
6/	3.4.2.2	4.8.2	Voltage regulation
7	3.4.11	4.17	Fungus resistance (90 days)
7/	3.4.3	4.9.1	Rated output
7/	3.4.2.2	4.8.2	Voltage regulation

1/, 2/, 3/, 4/, 5/, 6/, and 7/ Samples shall undergo simultaneously those tests marked with same footnote number.

4.3.1 Generating system. Seven generating systems (see 3.1) shall be identified for first article inspection and shall undergo the examinations and tests as specified in 4.3.

4.3.2 Repair parts. Seven samples of an individual part shall be substituted in generating systems previously approved as having undergone first article inspection (see 4.3). Subsequently, generating systems shall undergo examinations and tests specified in 4.3.

4.4 Conformance inspection. Conformance inspection shall include the examinations of 4.4.2 and the tests of 4.4.3.

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4.4.1 Sampling. Sampling plan for an inspection lot shall be as specified in the contract or order (see 6.2).

4.4.2 Examinations. Generating system and repair parts, shall be examined for the defects in table II. Any redesign or modification of the contractor's standard product to comply with specified requirements shall receive particular attention for adequacy and suitability. This element of inspection shall encompass all visual examinations and dimensional measurements. Noncompliance with any specified requirements or presence of one or more defects preventing or lessening maximum efficiency shall constitute cause for rejection.

4.4.3 100 percent (%) inspection. All generating systems and repair parts, of one type or part number, from an identifiable production period, from one manufacturer, shall be tested in the order listed in accordance with table III, stabilized frame temperature will not be required.

TABLE II. Classification of defects.

Category	Defect	Method of inspection
Major:		
101	Dimensions affecting interchangeability not within tolerance (see 3.3 & 3.3.1).	SIE 1/
102	Improper polarity (see 3.3.2).	SIE
103	If safety conditions are not as specified (see 3.7).	Visual & Functional
Minor:		
201	Dimensions not affecting interchangeability not within tolerance (see 3.3 & 3.3.1).	SIE
202	Improper finish (see 3.5).	Visual
203	Improper marking (see 3.6).	Visual

1/ SIE = Standard Inspection Equipment.

TABLE III. Conformance inspections.

Title	Requirement	Test
Rated output	3.4.3	4.9.1.1
Voltage regulation	3.4.2.2	4.8.2.1
Non-operating current	3.4.3.1	4.9.2

4.5 Control tests.

4.5.1 Sampling. Control test samples shall be selected from production lot which has passed conformance examination as specified in 4.4.2 and acceptance tests as specified in 4.4.3.

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Control test samples shall be selected at the rate of one generator system for each 250 produced except that not more than two shall be selected in any 30 day period.

4.5.2 Procedure. Control test samples shall be identified as to production period and subjected to test specified in table IV.

TABLE IV. Order of control tests.

Test	Requirement	Test
Overspeed	3.4.5	4.11
Voltage regulation	3.4.2.2	4.8.2
Submersion	3.4.10	4.16
Transient voltage	3.4.2.3	4.8.3 at 77°F ± 15°F only

4.6 Method of inspection.

4.6.1 Materials and design. Conformance to 3.2, 3.2.1, 3.2.2, 3.3, and 3.3.3 shall be determined by inspection of contractor records providing proof or certification that materials and design conform to requirements. Applicable records shall include drawings, specifications, design data, receiving inspection records, processing and quality control standards, vendor catalogs and certifications, industry standards, test reports, and rating data.

4.6.2 Defects. Conformance to 3.3.1, 3.3.2, 3.5, 3.6, 3.6.1, 3.6.2, and 3.7 shall be determined by examination for the defects listed in table II. Examinations shall be visual, tactile, or by measurement with Standard Inspection Equipment (SIE).

4.7 Electromagnetic interference characteristics test. To determine conformance to 3.4.1, generating system shall be subjected to test methods CE 102 and RE 102 of MIL-STD-461.

4.8 Voltage test. To determine conformance to 3.4.2, generating system shall undergo procedures specified in 4.8.1 through 4.8.3.2.

4.8.1 Voltage adjustment test. To determine conformance to 3.4.2.1, generating system shall be connected in circuit as shown in figure 3 and subjected to the following procedure:

a. At generator speed of 2400 rpm, using battery load only, adjust regulator potentiometer to maximum and minimum and record voltage.
b. Increase load to 60 A, repeat adjustments, and record voltage as in a.
c. Increase generator speed to 8000 rpm and repeat a and b.

4.8.2 Voltage regulation test. To determine conformance to 3.4.2.2, voltages shall be recorded with generating system operating under following conditions:

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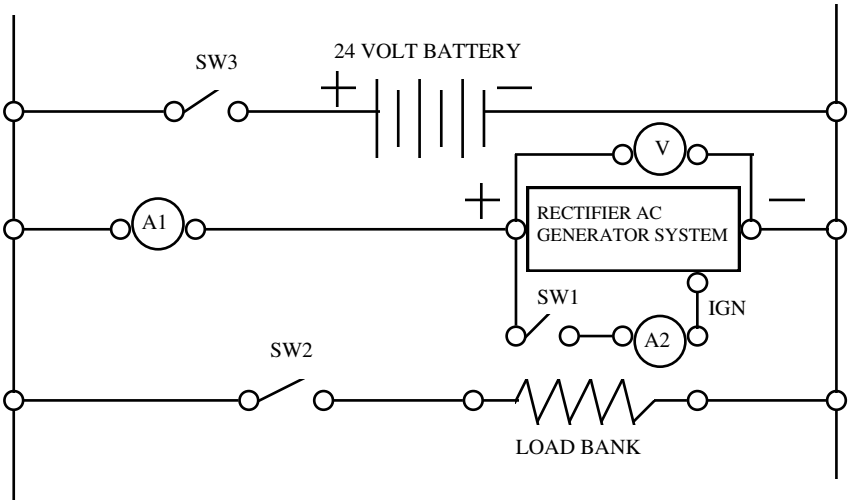
	RPM	A
a.	2500 ± 50	0-4.5
	2500 ± 50	60
	2500 ± 50	0-4.5
	2500 ± 50	30
	2500 ± 50	0-4.5
	2500 ± 50	45
b.	6000 ± 120	Same sequence as in a.
c.	10 000 ± 200	Same as in a.

NOTE: Change in load resistance shall be accomplished in 10 milliseconds (ms) maximum time.

4.8.2.1 Voltage regulation - acceptance test. All generating systems furnished under this specification shall be operated in ambient air temperature of 77 ± 15°F (25 ± 8.3°C) with frame temperature non-stabilized through the following cycle of speeds and loads:

	RPM	A
a.	2500 ± 50	0-4.5
b.	2500 ± 50	60
c.	10 000 ± 200	0-4.5
d.	10 000 ± 200	60

During this procedure, generating system shall conform to 3.4.2.2.



NOTE: A=AMMETER
AC=ALTERNATING CURRENT
IGN=IGNITION
SW=SWITCH
V=VOLTMETER

FIGURE 3. Generator system connected in circuit.

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4.8.3 Transient voltage characteristics test. To determine conformance to 3.4.2.3, generating system shall be subjected to test procedures specified in MIL-STD-1275, 4.8.3.1, and 4.8.3.2.

4.8.3.1 High voltage transients. The generating system shall be connected in test circuits as shown in figure 3 with SW2 and SW3 open and SW1 closed at zero rpm. A positive square wave voltage pulse of 100 V and 50 ms duration shall be applied to generating system output terminal once every two seconds for a period of 10 minutes (min). The generating system shall then be connected in a test circuit as shown in figure 4 with SW1 in position 1. Capacitor C shall be charged from power source, through R, until V1 reads 250 ± 5 V. SW1 shall then be changed from position 1 to position 2 within 1 s and held in that position for 15 s. This procedure shall be repeated ten times at 1 min intervals.

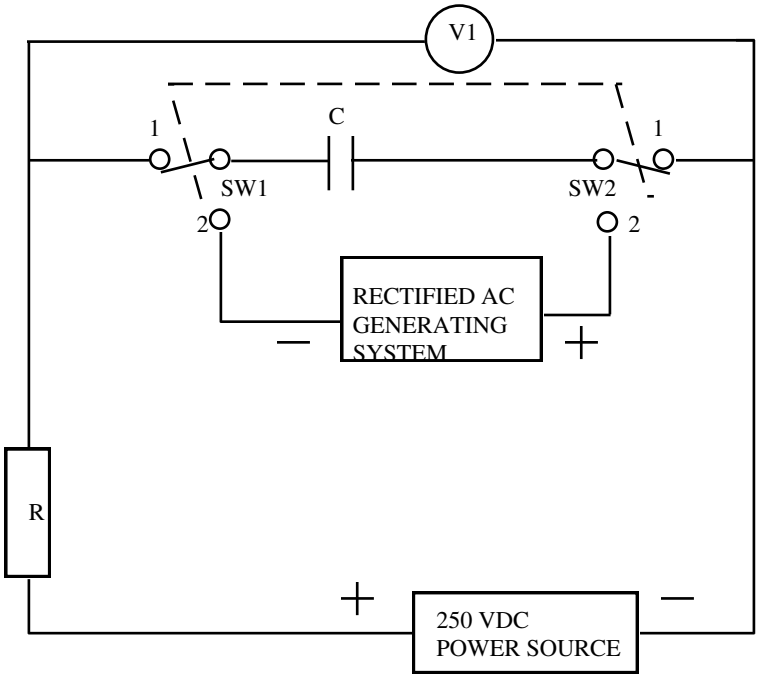
4.8.3.2 Load switching. Generating system (with voltage set at 28 ± 0.2 V) shall be operated at 6000 rpm when connected in a circuit as shown in figure 3, during 150 cycles of following test procedure:

1) Close switch 1.
2) Close switch 3 until system voltage reaches “set voltage”.
3) Close switch 2 to apply a 60 A resistance load.
4) Open switch 3 after load is applied.
5) Open switch 2 while generating system is producing 60 A.
6) Wait until voltage decreases to within 10% of “set voltage”.

The test shall be accomplished in not more than 15 min at each of the following temperatures:

a. -65° ± 5°F (-54 ± 2.7°C).
b. 77° ± 15°F (25 ± 8.3°C).

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NOTE:
AC= ALTERNATING CURRENT
C= CAPACITOR= 30 MICROFARAD (μf)
R= RESISTOR= 200 KILOHM (KΩ)
SW= SWITCH
ALL WIRE TO BE 10 AMERICAN WIRE GAGE (AWG) OR HEAVIER

FIGURE 4. Negative transient voltage test circuit.

4.9 Current tests. To determine conformance to 3.4.3, generating system shall undergo procedures specified in 4.9.1 through 4.9.2.

4.9.1 Rated output test. To determine conformance to 3.4.3, unless otherwise specified in applicable drawing, regulator voltage sensing lead shall be connected to same point at which voltage is measured. The following test shall be performed:

a. Connect generating system in circuit as shown in figure 3.
b. Unless otherwise specified herein, battery shall remain connected for all tests and switch SW3 shall be closed.
c. Speed, frame temperature, ignition lead current, output current, and system voltage shall be recorded.
d. Voltage adjustment screw shall be rotated to stop on high voltage side of adjustment range.

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e. Generating system speed and load shall be adjusted to maximum load for each point on speed-load graph represented by chart shown on applicable drawing with voltage maintained at 28 ± 0.2 V.
f. Temperature of generating system shall be stabilized (see 4.2.2) at each point on performance curve represented by chart shown on applicable drawing.

4.9.1.1 Rated output - acceptance test. All generating systems furnished under this specification shall be connected into circuit shown in figure 3 with battery. Speed shall be adjusted to not more than 1800 rpm. Load shall be adjusted to not less than 60 A. Generating system shall be so operated for not less than one minute and shall give evidence of conformance to 3.4.3. Stabilized frame temperature will not be required.

4.9.2 Non-operating current test. To determine conformance to 3.4.3.1, generating system shall be connected in circuit specified in figure 3. With switches SW3 and SW1 closed, and SW2 open, non-operating steady state current measured by ammeter A2 shall be ≤ 300 mA. Subsequently, switch SW1 shall be opened and current measured by ammeter A1 shall be ≤ 25 mA.

4.10 Endurance test. To determine conformance to 3.4.4, generating system shall be connected in circuit as specified in figure 3 and tested in accordance with following procedure:

a. Set regulator at 28 ± 0.2 V (set voltage).
b. Adjust load for 60 A.
c. Operate generating system for a total of 1000 hours (hr) in accordance with speed and duration indicated in table V, recording frame temperature, generating system voltage, and generating system current.

TABLE V. Endurance test - sequence.

RPM	Duration	Generating system position
2500	336 hr	Shaft horizontal
10 000	166 hr	Shaft horizontal
6000	166 hr	Shaft horizontal 1/
6000	166 hr	Shaft 45° ± 1° above horizontal
6000	166 hr	Shaft 45° ± 1° below horizontal

1/ Throughout this test period, load shall be cycled from 0-5 A to 60 A (30 min at each load).

Sequence indicated in table V shall be conducted three times for a total test time of 3000 hr.

NOTE: Throughout endurance test, belt loading shall be maintained in accordance with 6.4.3.

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4.11 Overspeed test. To determine conformance to 3.4.5, generating system shall be connected in a circuit as specified in figure 3. With 60 A load, generating system shall be operated at 10 000 rpm until frame temperature is stabilized (see 4.2.2). Subsequently, load shall be reduced to less than 5 A and speed increased to 12 000 rpm. Speed and load shall continue for 15 seconds.

4.12 High temperature test. To determine conformance to 3.4.6, generating system shall be connected in circuit as specified in figure 3 in ambient air temperature of 250°F (121°C). With maximum load attainable at 28 ± 0.2 Vdc and 1750 ± 50 rpm, generating system shall be operated until frame temperature is stabilized (see 4.2.2). Subsequently, load shall be adjusted to 5 A less than maximum load at which system must function continuously to maintain output voltage. This load shall be not less than 39 A. Generating system shall be so operated for 200 hr in ambient air temperature of 250°F (121°C). Air movement in high temperature chamber shall be held to minimum to prevent temperature gradient within chamber and to prevent cooling from excessive air motion. Air motion from generating system shall not be considered in evaluating conformance. Subsequently, generating system shall undergo test procedure specified in 4.9 in ambient air temperature specified in 4.2.

4.13 Low temperature test. To determine conformance to 3.4.7, generating system shall undergo the following test:

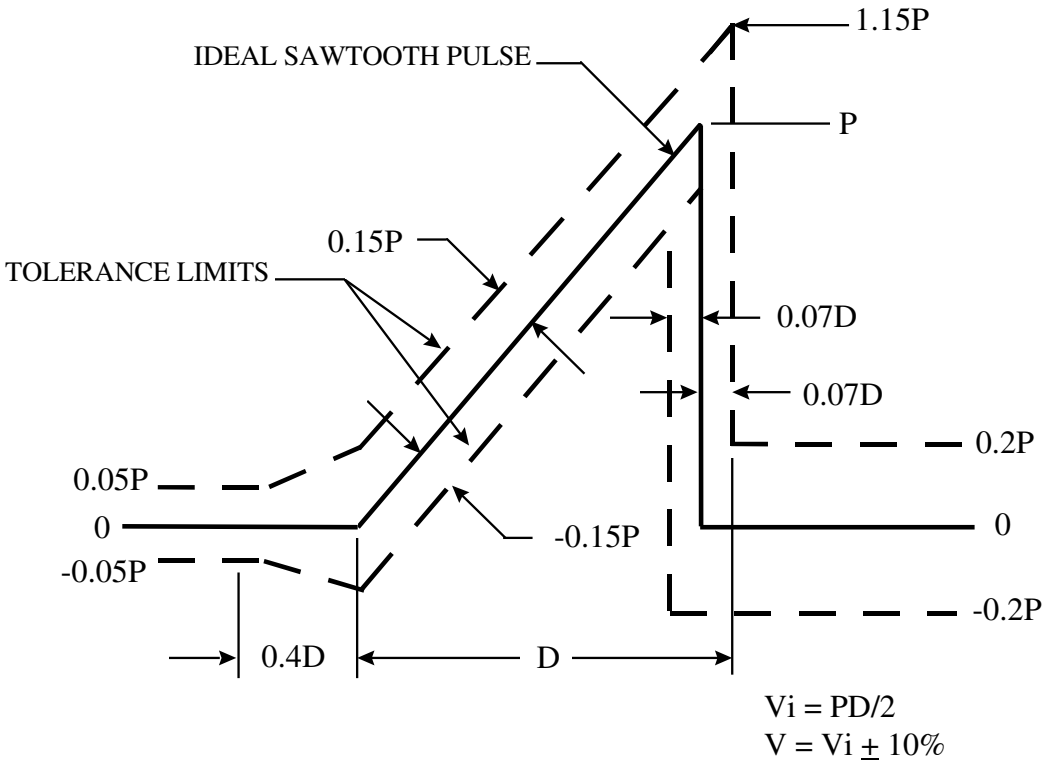
a. Prior to testing, generating system shall be stored at temperature of -80°F (-59°C) for a period of 24 hr.
b. After removal from storage, generating system frame temperature shall be stabilized in chamber with ambient temperature of -65°F (-54°C).
c. Generating system shall be operated at 1750 ± 50 rpm with 60 A load for a period of 4 hr.
d. During test, voltage shall remain at set voltage ± 0.5 V.

Subsequently, system shall undergo test procedure specified in 4.9 in ambient air temperature specified in 4.2.

4.14 Shock. To determine conformance to 3.4.8, the generator shall be subjected to the sawtooth shock pulse form shown in figure 5 herein. The shock machine may be of the free fall, resilient, rebound, non-resilient, hydraulic, compressed gas, or other activating types. The generator shall be mounted on a cradle for the testing, whenever possible, the test load shall be distributed uniformly on the test platform in order to minimize the effects of unbalanced loads. There shall be three shocks in each direction applied along the three mutually perpendicular, axes of the generator (18 shocks). They shall have a peak value 100 g's and a nominal duration (D) of 6 ms in accordance with figure 5. The velocity change (Vi) shall be 9.7 feet per second (ft/sec).

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Measurements are to be made before and after the shocks unless otherwise specified, and during the test if specified (see 6.2). Subsequently, the generator shall be subjected to the test specified in 4.8 and 4.9.



NOTE: The oscillogram should include a time about 3D long with the pulse approximately in the center. The integration to determine the velocity change should extend from .4D before the pulse to .1D beyond the pulse. The peak acceleration magnitude of the sawtooth pulse is P and its duration is D. Any measured acceleration pulse which can be contained between the broken line boundaries is a nominal terminal-peak sawtooth pulse of nominal peak value, P, and nominal duration, D. The velocity-change associated with the measured acceleration pulse is V.

FIGURE 5. Tolerances for terminal-peak sawtooth shock pulse.

4.15 Vibration. To determine conformance to 3.4.9, generating system shall be tested in accordance with test condition specified herein. The following conditions shall apply:

a. Frequency range shall be 10 to 600 cycles per second (Hertz (Hz)).
b. Maximum load along longitudinal (shaft) axis shall be 10 gravity units (g) applied at vibration table.

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c. Alternator shaft shall be rotated throughout vibration test at speed of 1750 ± 50 rpm.
d. Alternator shall be rigidly mounted as in normal vehicle operating position.

The generator, while deenergized or operating under the load condition specified, shall be mounted as specified using suitable mounting apparatus to assure that mounting is free from resonance over the test frequency range, and shall be subjected to a simple harmonic motion having an amplitude of either 0.06 in double amplitude (maximum total excursion) or G=10 g(peak) ($G = .512\ f^2DA$, where f is the frequency in Hz, DA is the double amplitude in inches), which ever is less. The tolerance on vibration amplitude shall be ± 10%. The vibration frequency shall be varied logarithmically between the approximate limits of 10 to 600 Hz and return to 10 Hz. The frequency range shall be traversed in approximately 20 min. This cycle shall be performed 12 times in each of three mutually perpendicular directions (total of 36 times), so that the motion shall be applied for a total period of approximately 12 hr. Interruptions are permitted provided the requirements for rate of change and test duration are met. Completion of cycling within any separate band is permissible before going to the next band.

Subsequently, system shall undergo test procedures specified in 4.10.

4.16 Submersion. To determine conformance to 3.4.10, generating system shall be frame temperature stabilized (see 4.2.2) by operating at 28 V, 2500 ± 50 rpm and load current of 60 A. Stabilization shall be accomplished in dry condition. Generating system shall then be subjected to following test cycle:

a. Generating system shall be driven in dry condition for a period of 60 hr at 7000 ± 100 rpm and no load.
b. Alternator shaft shall then be rotated for a period of eight hours while completely submerged in a 5% (by weight) salt solution at a speed of 1750 ± 50 rpm. At end of first hour only, rotation shall be stopped and battery current measured while unit remains submerged. Switches SW1 and SW3 shall be closed and switch SW2 open for a and for first part only of b when connected as shown in figure 3. During last seven hours of b, all switches shall be open.
c. Subsequent to submersion, regulator compartment shall be opened to determine whether solution is present and amount of solution found therein.
d. Generating system shall undergo test procedures specified in 4.8 and 4.9.

4.17 Fungus resistance. To determine conformance to 3.4.11, system shall be tested in accordance with method 508.6 of MIL-STD-810 except that test period shall be a continuous 90 days. Upon removal from cabinet, examination shall be accomplished without disturbing any fungus growth. Upon completion of examination and replacement of access port covers, generating system shall be operated for one hour without load at 6500 rpm. Subsequently, generating system shall undergo test procedures specified in 4.8 and 4.9.

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4.18 Corrosion resistance. To determine conformance to 3.4.12, generating system shall be tested for resistance to corrosion in accordance with ASTM B117, and the length of test shall be 200 hr. During test, generating system will be rotated for a ten minute period out of each 8 hr test interval. After test, alternator shaft shall be rotated for 10 minutes at 1750 ± 50 rpm and no load. Subsequently, generating system shall undergo test procedures specified in 4.8 and 4.9.

4.19 Sand and dust test. To determine conformance to 3.4.13, generating system shall be subjected to method 510.3, sand and dust test of MIL-STD-810. Dust velocity shall be 2300 ± 500 feet per minute (ft/min). During 10 min of each hour of test period, system shall be driven at a speed of 3450 ± 50 rpm.

5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When actual packaging of materiel is to be performed by DoD personnel, these personnel need to contact the responsible packaging activity to ascertain requisite packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activity within the Military Department or Defense Agency, or within the Military Department's System Command. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

6. NOTES

(This section contains information of a general or explanatory nature which may be helpful, but is not mandatory.)

6.1 Intended use. Generating systems covered by this specification are intended for use in the high mobility multipurpose wheeled vehicle (HMMWV), 2-1/2 and 5 ton trucks, M939, and assorted construction equipment vehicles. The generating systems are used to produce direct current electrical energy for various requirements, such as lighting and battery charging, under all conditions of climate, terrain, and military service.

6.2 Acquisition requirements. Acquisition documents must specify the following:

a. Title, number, revision, and date of this specification.
b. Title, number, revision, and date of part drawing.
c. If required, the specific issue of individual documents referenced (see 2.2.1 and 2.3).
d. First article inspection, if required (see 3.1).
e. If finish is other than as specified (see 3.5).
f. Marking if other than as specified (see 3.6.1).
g. Inspection conditions if other than as specified (see 4.2).

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h. Sampling plan requirements (see 4.4.1).
i. Packaging requirements (see 5.1).

6.3 Outercoat. Master Painter Institute (MPI) #9 – Alkyd, Exterior, Gloss, (MPI Gloss Level 6) or MPI #94 – Alkyd, Exterior, Semi-gloss, (MPI Gloss Level 5) may be used on primed exterior metal.

6.4 Definitions.

6.4.1 Set voltage. As used in this specification, "set voltage" is defined as the output voltage to which the generator system is adjusted.

6.4.2 Stabilized temperature. Frame temperature should be considered stabilized when it varies not more than 8°F (4.5°C) between high and low readings in any 1 hr of operation under constant conditions.

6.4.3 Belt loading. Maximum dynamic radial shaft loading to obtain bid bearing life at various alternator speeds computed with the following and table VI:

$$L = \frac{F}{8.46 + X}$$

L = Resultant of maximum radial shaft load applied by belt drive at operating condition in pounds.
X = Distance from inboard face of pulley hub to centerline of belts in inches.
F = load factor

TABLE VI. Load factor.

ALTERNATOR RPM	BID BEARING LIFE IN HRS		
	1000	2000	3000
	F	F	F
1000	9350	7300	6300
2000	7350	5900	5100
3000	6250	5100	4400
4000	5600	4570	4000
5000	5150	4250	3700
6000	4850	4000	3500
7000	4620	3800	3300
8000	4450	3650	3170
9000	4300	3550	3050
10 000	4200	3450	2950

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6.4.4 Tin whisker growth. The use of alloys with tin content greater than 97 percent may exhibit tin whisker growth problems after manufacture. Tin whiskers may occur anytime from a day to years after manufacture, and can develop under typical operating conditions on products that use such materials. Conformal coatings applied over top of a whisker-prone surface will not prevent the formation of tin whiskers. Alloys of 3 percent lead have shown to inhibit the growth of tin whiskers.

6.5 Subject term (key word) listing.

Battery charging
Charging
Current
Generator
HMMWV
Lighting
M939
Voltage

6.6 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.

Custodian:
Army - AT
DLA - CC

Preparing Activity:
Army – AT

(Project 2920-2009-005)

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